

2026 FALL SEMESTER
Programming for Data Science Lab (MACSE502) Record
INDEX

Week No	Date	Sl. No.	Type	Topic	Title with download Link for document	Problem Statement	Page No.
1	09-07-2026	1	PP	Pandas	Basic Data Manipulation with Pandas	Create a DataFrame with columns Name, Age, and City containing data for five individuals. Perform the following operations: select only the Name and Age columns, filter rows where Age is greater than 25, add a new column Country with a default value, and sort the DataFrame by Age in descending order. Finally, calculate the average age of the individuals.	
1	09-07-2026	2	EP	Pandas			
2	16-07-2026	3	PP	Pandas	Data Cleaning and Preprocessing with Pandas	Create a DataFrame with some missing values in columns Name, Age, and City. Perform the following operations: fill missing values in the Age column with the mean age, drop rows where Name or City is missing, and convert the Age column to integer type. Finally, normalize the Age column using min-max scaling	
2	16-07-2026	4	EP	Pandas			
3	23-07-2026	5	PP	NumPy	Creating and Manipulating Arrays Using NumPy	Create a 2D array of shape (3, 4) with random integers between 0 and 10. Perform the following operations: calculate the mean and standard deviation of the entire array, slice the array to get the first two rows and last two columns, reshape the array to shape (4, 3), and perform element-wise multiplication with another array of the same shape.	
3	23-07-2026	6	EP	NumPy			
4	30-07-2026	7	PP	Pandas NumPy Matplotlib	Feature Engineering and Exploratory Data Analysis	Feature Engineering, Exploratory Data Analysis: Create a DataFrame with columns Feature1, Feature2, and Target containing random data. Perform the following operations: create a new feature that is the logarithm of Feature1, bin Feature2 into three categories (low, medium, high), and calculate the correlation matrix of the DataFrame. Finally, create a scatter plot of Feature1 vs. Feature2 colored by Target, and interpret any visible patterns	

2026 FALL SEMESTER
Programming for Data Science Lab (MACSE502) Record
INDEX

Week No	Date	Sl. No.	Type	Topic	Title with download Link for document	Problem Statement	Page No.
4	30-07-2026	8	EP	Pandas NumPy Matplotlib			
5	06-08-2026	9	PP	Web Scraping	requests, BeautifulS ouppanda sjson	Scrape data from a webpage and store it in a structured format like CSV orJSON	
5	06-08-2026	10	EP	Web Scraping	requests, BeautifulS oup pandas json		
6	20-08-2026	11	PP	Plotly	Interactive Bar Chart with Plotly	Create an interactive bar chart showing the population of different cities	
6	20-08-2026	12	PP	Plotly	Interactive Scatter Plot with Plotly	Create an interactive scatter plot showing the relationship between house size and price	
7	27-08-2026	13	PP	Bokeh	Interactive Scatter Plot with Bokeh	Create an interactive scatter plot showing the relationship between petal length and petal width from the Iris dataset using Bokeh	
7	27-08-2026	14	EP	Bokeh	Interactive Line Plot with Bokeh	Create an interactive line plot showing the daily temperatures over a week using Bokeh.	
7	27-08-2026	15	EP	Bokeh	Interactive Bar Chart with Bokeh	Create an interactive bar chart showing the sales figures of different products using Bokeh	
8	03-09-2026	16	PP	scikit-learn and Plotly	Visualizing Clusters with scikit-learn and Plotly	Perform Principal Component Analysis (PCA) on the Iris dataset and visualize the first two principal components using an interactive 2D scatter plot in Plotly	
8	03-09-2026	17	EP	scikit-learn and Plotly	Visualizing Clusters with scikit-learn and Plotly	Perform K-means clustering on the Iris dataset and visualize the clusters using an interactive 3D scatter plot in Plotly	

2026 FALL SEMESTER
Programming for Data Science Lab (MACSE502) Record
INDEX

[illegible]